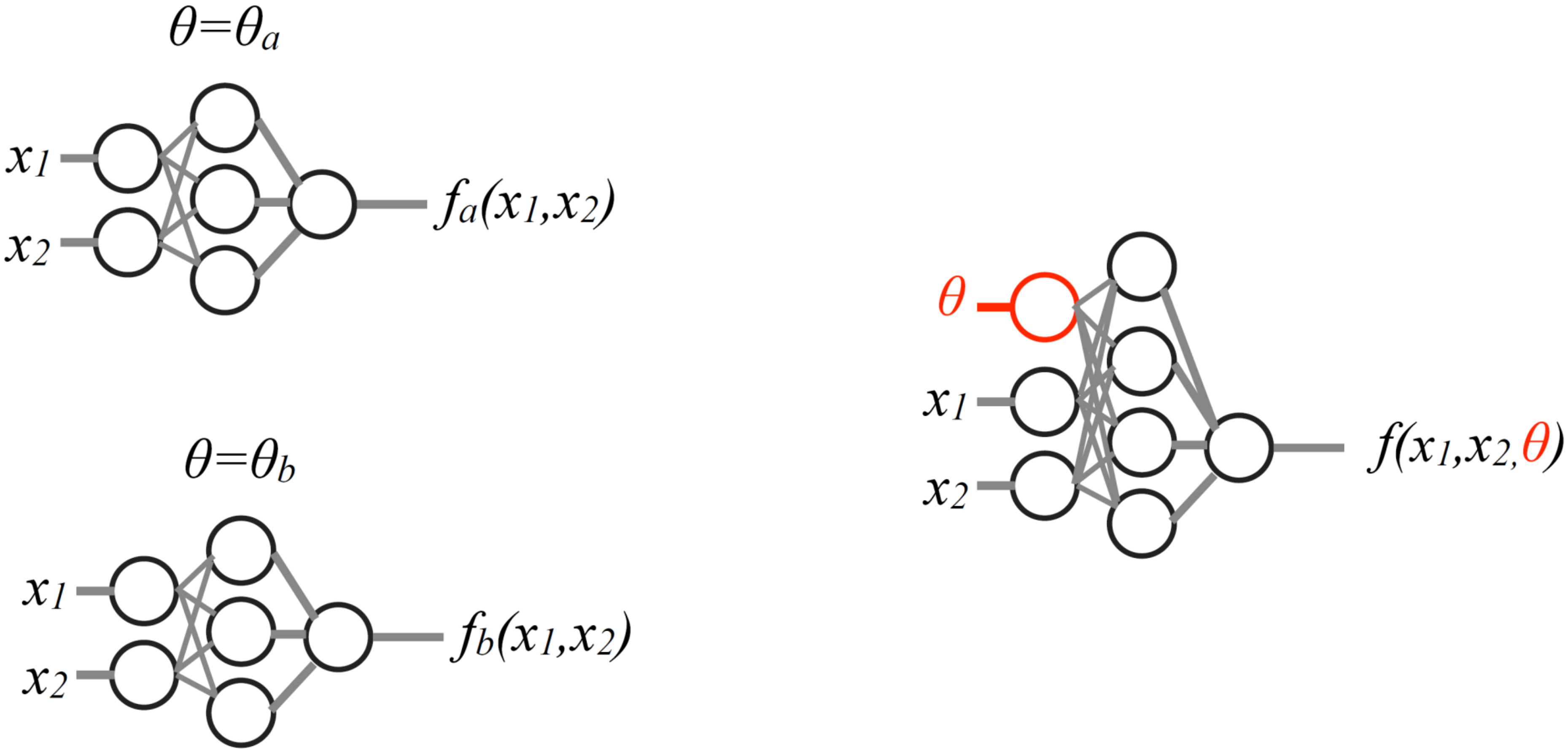
Prior Dependence of Parameterized Neural Networks

Parameterized Neural Networks



Baldi, P., Cranmer, K., Faucett, T. et al. Parameterized neural networks for high-energy physics. Eur. Phys. J. C 76, 235 (2016). https://doi.org/10.1140/epjc/s10052-016-4099-4

Theory

- training data: parameter-dependent and –independent data

$Y_0 = 0,\ X_0 = \left(x, \tilde{\theta}\right) \sim \left(P(x), P\left(\tilde{\theta}\right)\right)$ where $P(x) = \int p\left(x \middle| \hat{\theta}\right) p\left(\hat{\theta}\right) d\theta$

$Y_1 = 1,\ X_1 = (x, \theta) \sim \left(P(x|\theta), P(\theta)\right) = P(x, \theta)$

- classifier trained with Binary Cross Entropy Loss (BCE); if ideal:

$$\frac{f(x,\theta)}{1 - f(x,\theta)} = r(x,\theta) = \frac{p(x,\theta)}{p(x)p(\theta)} = \frac{p(\theta|x)}{p(\theta)}$$

- posterior distribution from “likelihood ratio trick“:

$$p(\theta|x) = \frac{f(x,\theta)}{1 - f(x,\theta)} p(\theta)$$

Toy Example

- training data from normal distribution:

for X_0: $p(x|\hat{\theta}) = \mathcal{N}(\mu = \hat{\theta}, \sigma = 1)$

for X_1: $p(x|\theta) = \mathcal{N}(\mu = \theta, \sigma = 1)$

- model architecture: 5-layer MLP
- multiple classifiers trained for different prior distributions:

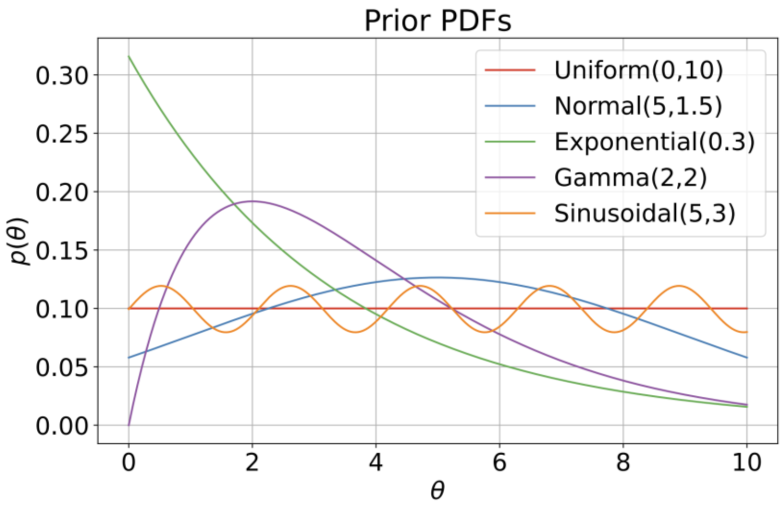


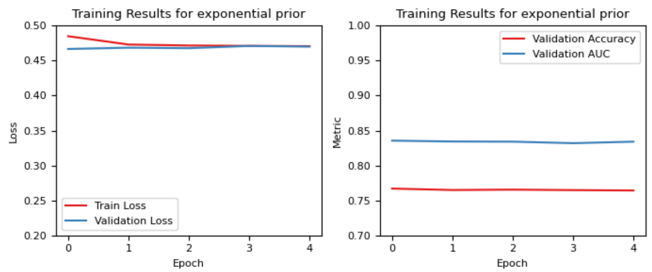
Figure 1: normalized prior distributions

Name	$p(\theta)$
Uniform	1
Normal	$\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
Exponential	$\exp(-\lambda x)$
Gamma	$x^{\alpha-1}\exp(-x/\theta)$
Sinusoidal	$\sin(\omega x) + b$

Table 3: unnormalized prior distributions

First Results

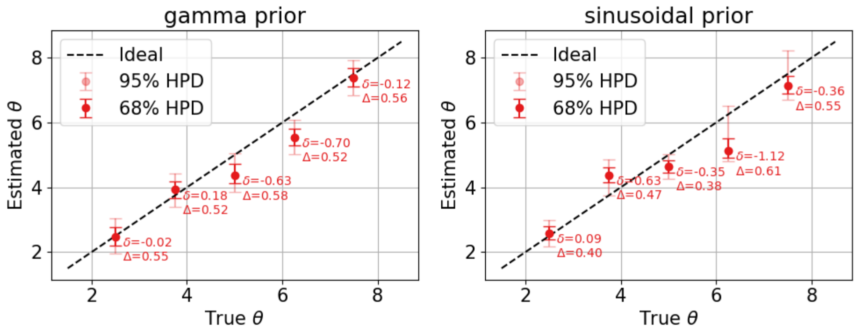
- converged training:

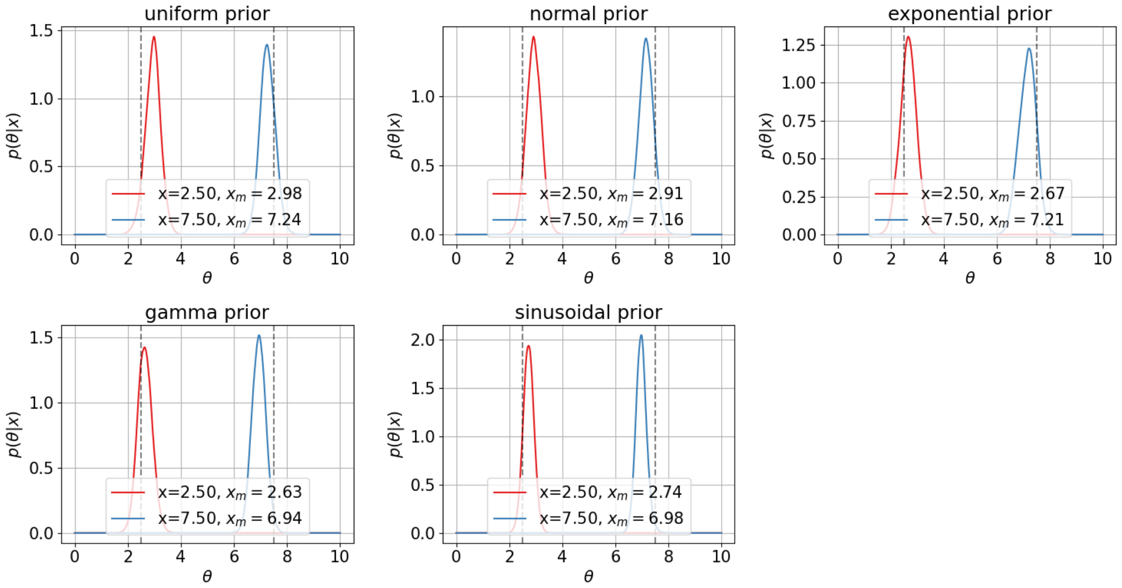


- worse performance for exponential & gamma prior:

Prior	Test Loss	Test Accuracy	Test AUC
Uniform	0.4035	0.8137	0.8694
Normal	0.4322	0.7955	0.8559
Exponential	0.4747	0.7603	0.7603
Gamma	0.4626	0.7715	0.8376
Sinusoidal	0.4236	0.8018	0.8593

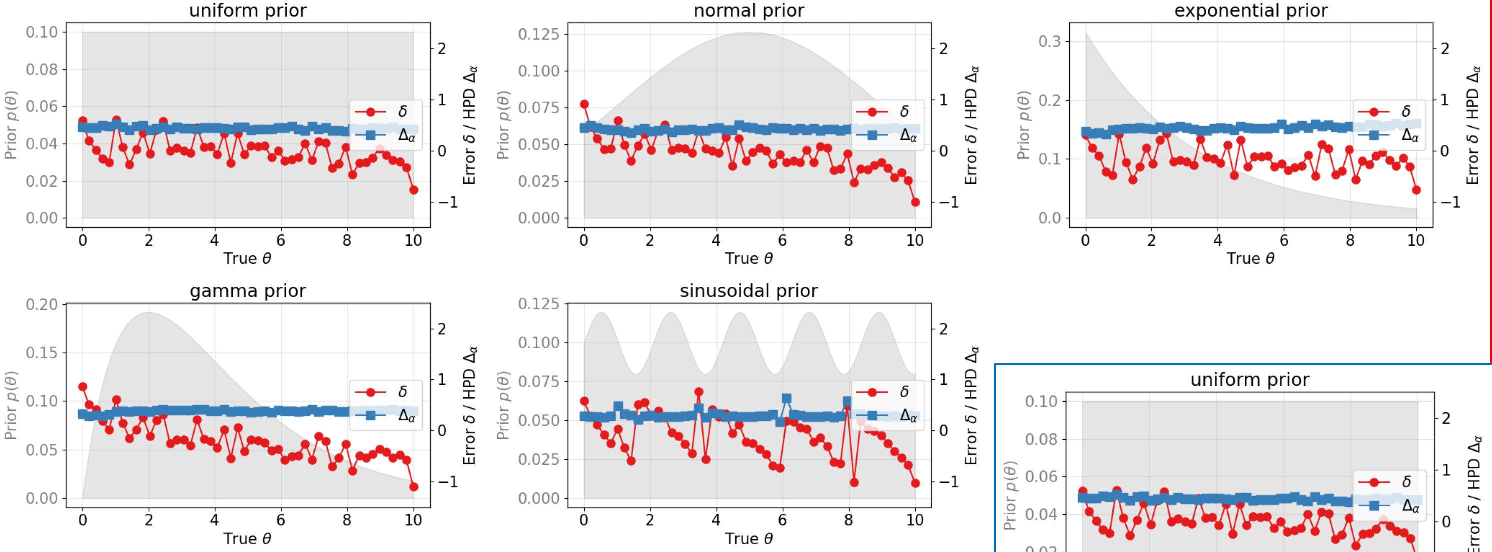
- posterior dependence on prior:





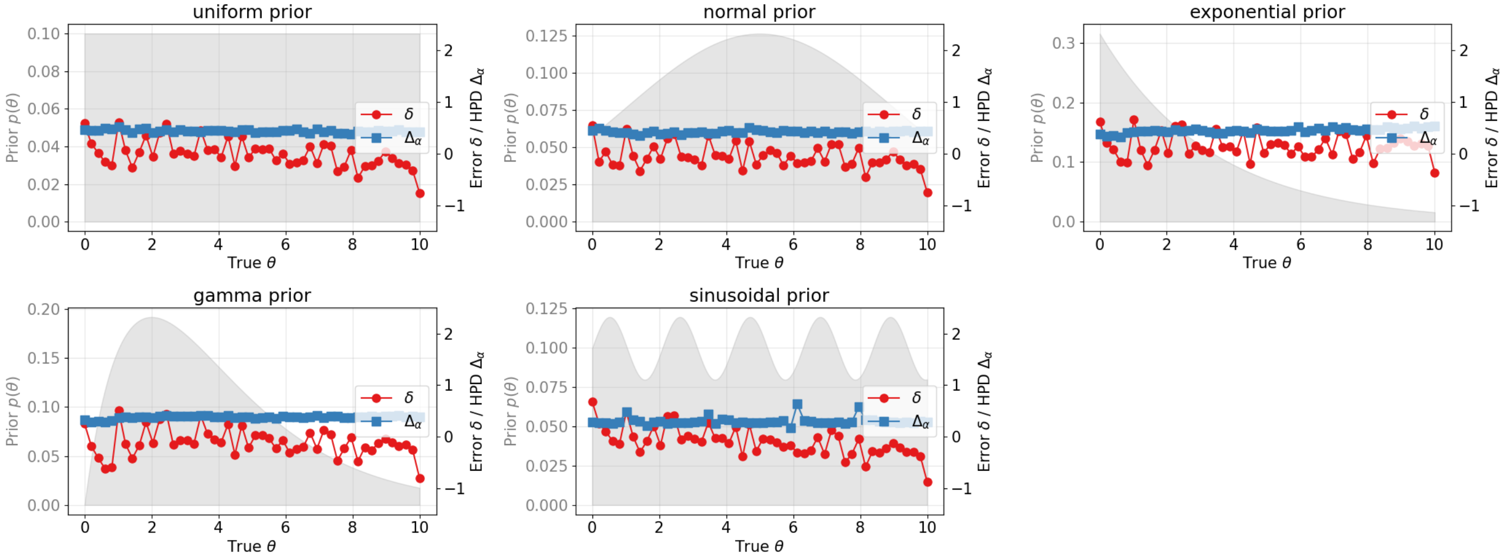
Problems with First Results

- Likelihood ratio shows less prior dependence on inference:



inference using posterior

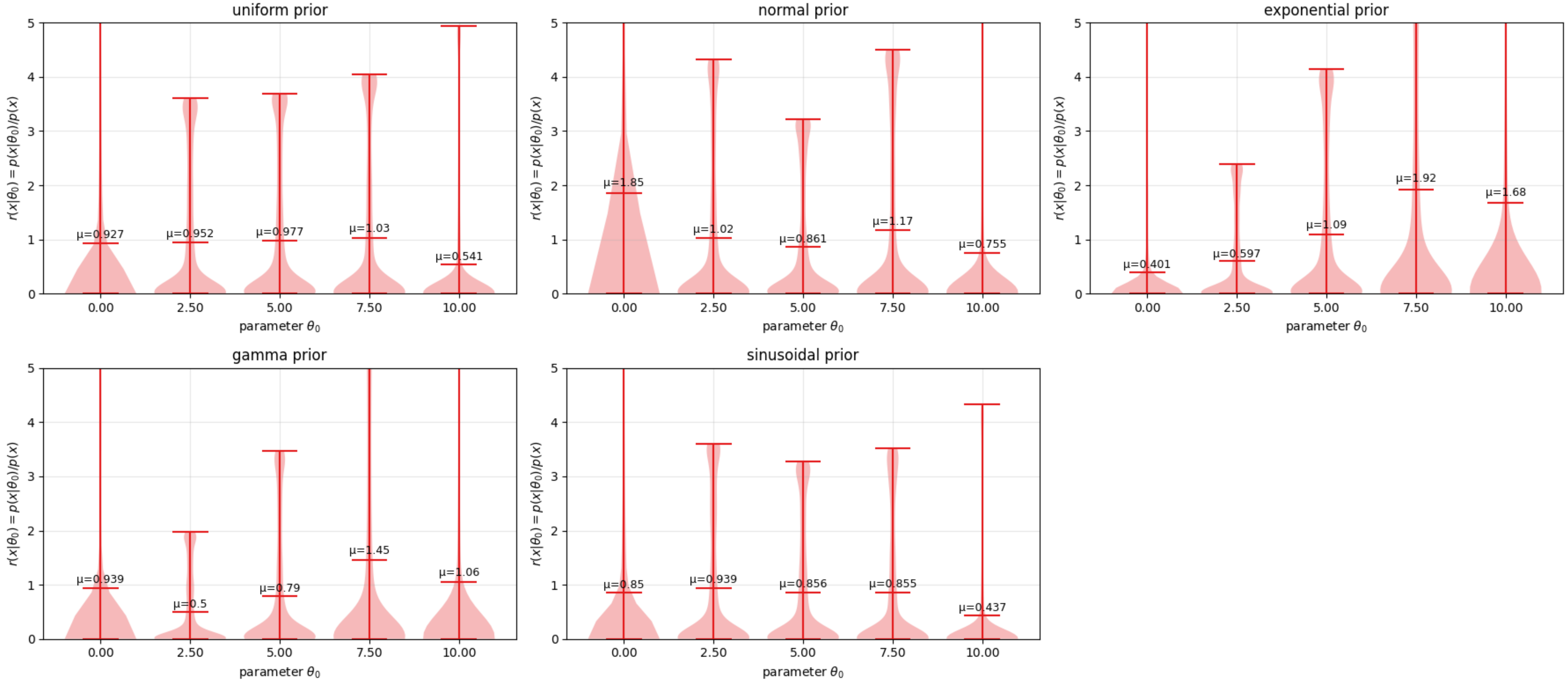
inference using likelihood ratio

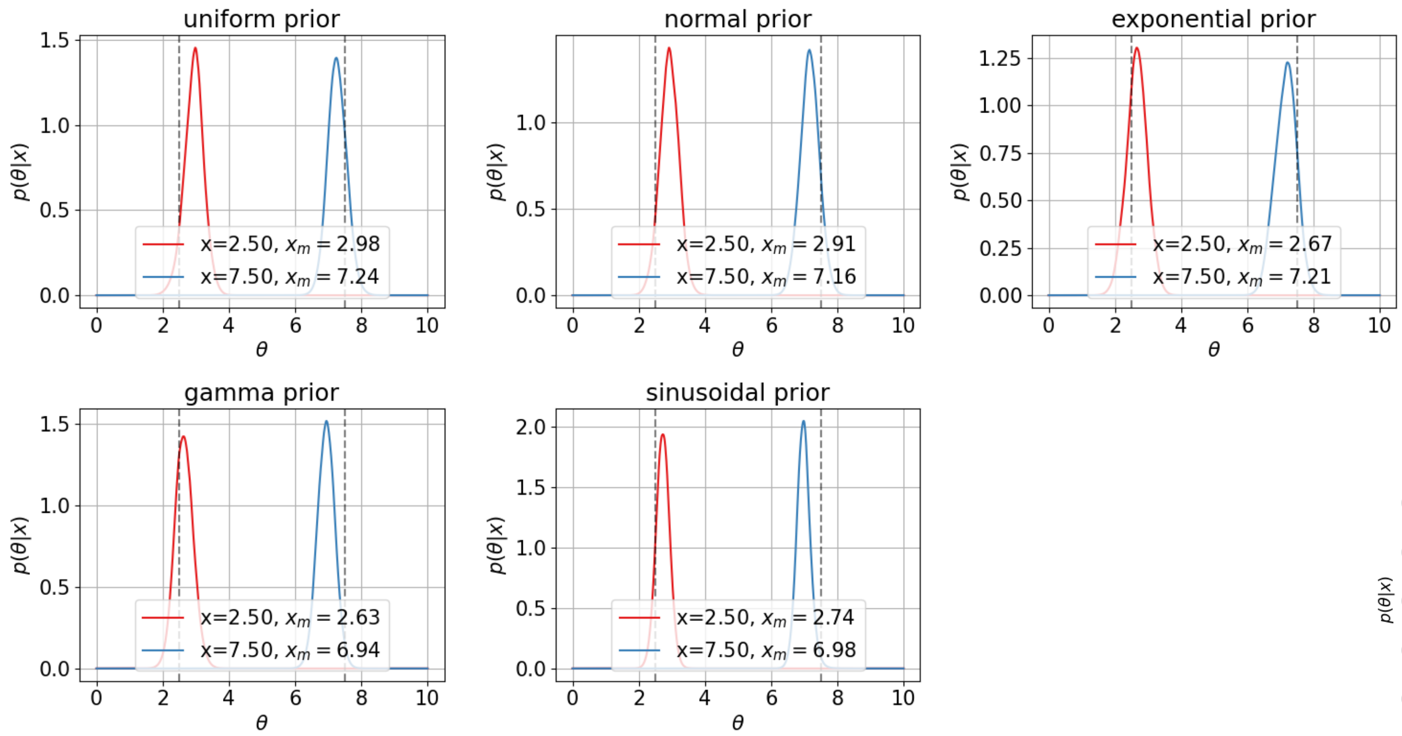


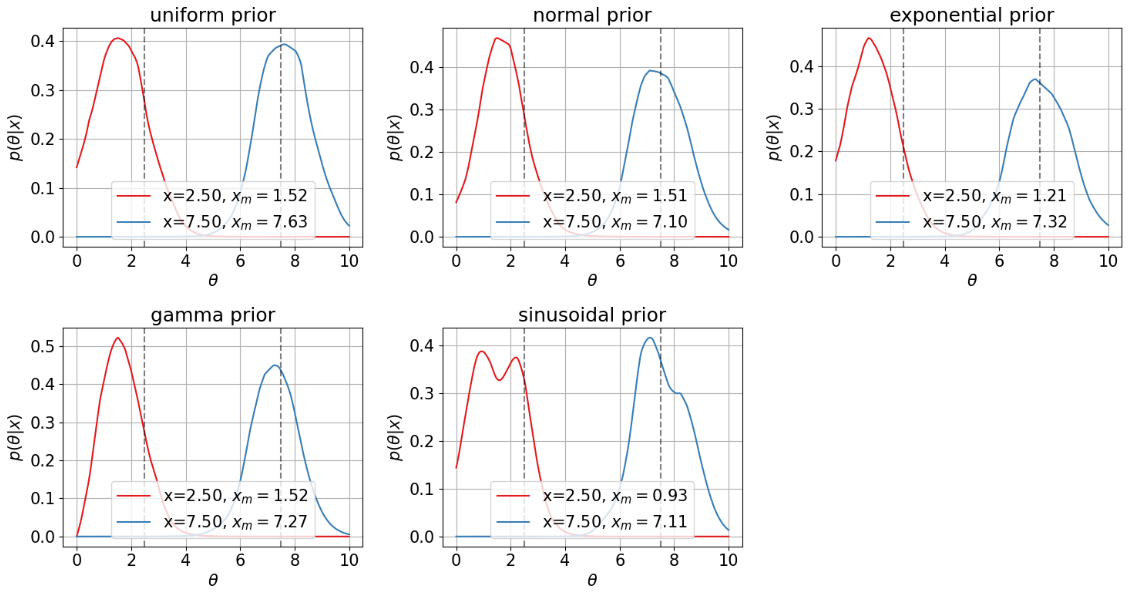
Posterior Verification

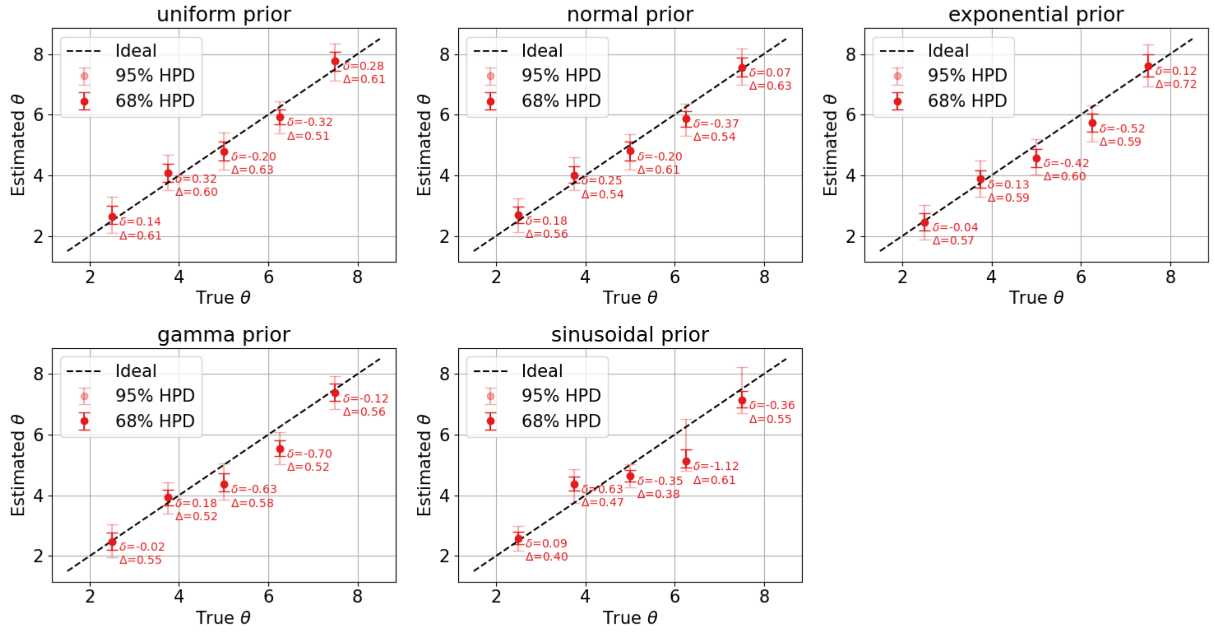
$$1 = \int p(x|\theta_0)dx = \int r(x|\theta_0)p(x)dx = \mathbb{E}[r(x|\theta_0)]$$

$$r(x,\theta) = \frac{p(x|\theta)}{p(x)} = \frac{p(\theta|x)}{p(\theta)}$$




uniform prior
x=2.50, $x_m = 2.98$
x=7.50, $x_m = 7.24$
normal prior
x=2.50, $x_m = 2.91$
x=7.50, $x_m = 7.16$
exponential prior
x=2.50, $x_m = 2.67$
x=7.50, $x_m = 7.21$
gamma prior
x=2.50, $x_m = 2.63$
x=7.50, $x_m = 6.94$
sinusoidal prior
x=2.50, $x_m = 2.74$
x=7.50, $x_m = 6.98$
$p(\theta|x)$
θ


uniform prior
x=2.50, $x_m = 1.52$
x=7.50, $x_m = 7.63$
normal prior
x=2.50, $x_m = 1.51$
x=7.50, $x_m = 7.10$
exponential prior
x=2.50, $x_m = 1.21$
x=7.50, $x_m = 7.32$
gamma prior
x=2.50, $x_m = 1.52$
x=7.50, $x_m = 7.27$
sinusoidal prior
x=2.50, $x_m = 0.93$
x=7.50, $x_m = 7.11$
$p(\theta|x)$
θ


uniform prior
normal prior
exponential prior
gamma prior
sinusoidal prior
Ideal
95% HPD
68% HPD
Estimated θ
True θ